UNIVERSITI SAINS MALAYSIA

School of Electrical and Electronics Engineering
Universiti Sains Malaysia, Engineering Campus

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EEM 441

Control & Instrumentation Laboratory

Experiment 10

INTERFACING FPGA TO A DIGITAL-TO-ANALOG CONVERTER

Date : ______________________________

Group No. : ______________________________

Group Members : ______________________________ Matrix # __________

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Experiment 10

Interfacing FPGA to Digital-to-Analog Converter

Objectives

1. To understand the operation of a digital-to-analog converter (DAC).
2. To interface an FPGA, via the Altera DE2 board, to a DAC.

Equipment / Software Required

1. Altera Quartus II software
    - version 9/10/12 for Cyclone II
    - version 9/10/12/latest for Cyclone IV
2. Altera DE2 board
3. Multimeter / oscilloscope
4. DAC0800 digital-to-analog converter
5. Resistors: $10\,\mathrm{k\Omega}$ ($\times 4$), $100\,\Omega$ ($\times 1$)
6. Capacitor: $0.01\,\mu$F

Introduction

A digital-to-analog converter (DAC) converts digital pulses into an analog signal. The first criterion for judging a DAC is its resolution, which depends on the number of binary inputs: the number of distinct analog output levels equals 2^n, where n is the number of data-bit inputs. An 8-bit DAC such as the DAC0800 therefore provides 256 discrete voltage (or current) output levels.

A DAC can be used to generate a sine wave. The output voltage for a given angle θ is:

$$V_{out} = 5\,\mathrm{V} + (5\,\mathrm{V} \times \sin\theta)$$

To find the digital value to send to the DAC for a given angle, multiply the resulting V_{out} by 25.6 (since there are 256 steps across a $10\,\mathrm{V}$ full-scale output).

Procedure

Activity 1 — DAC0800 operation

Objective: understand the operation of the DAC0800.

1. Connect the DAC0800 as shown in Figure 10.1.

2. Apply the digital input values at D0–D7 as listed in Table 10.1.
3. Measure the resulting analog output and tabulate your results in Table 10.1.

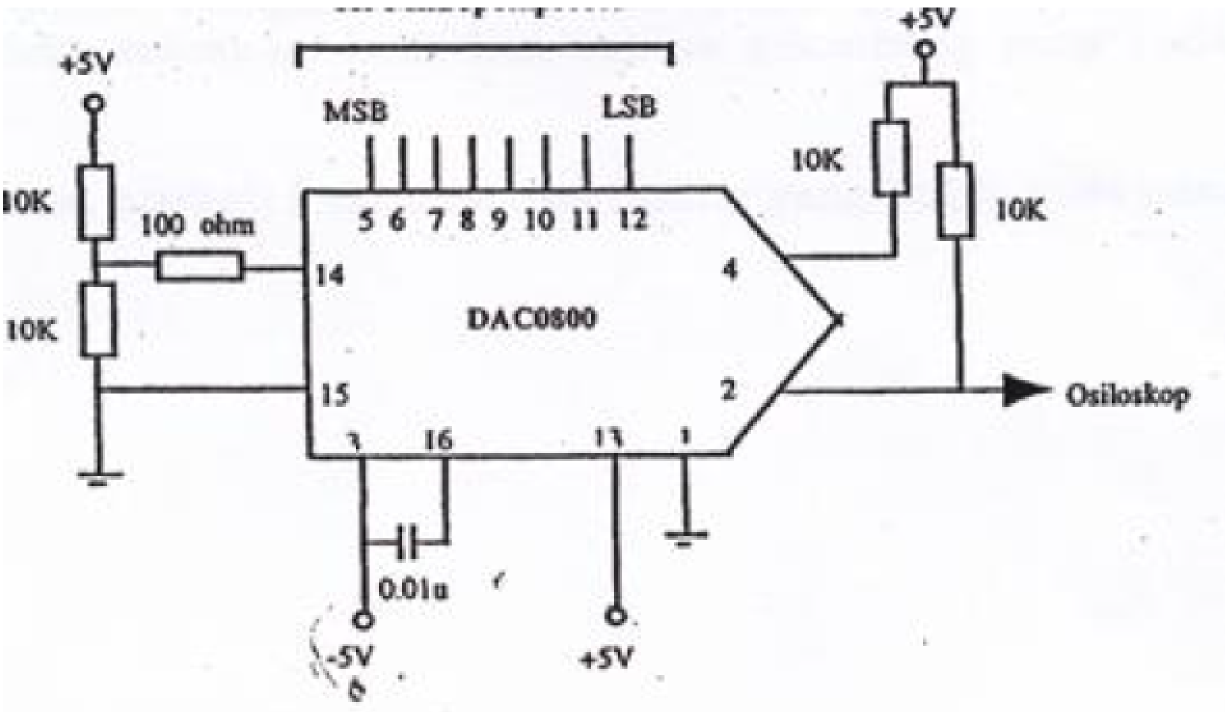

Figure 10.1: DAC0800 test circuit with op-amp output stage.

Table 10.1: DAC0800 digital-to-analog conversion results.

Digital value (decimal)	Analog value
0	
10	
60	
80	
100	
128	
180	
200	
220	
255	

Activity 2 — Generating a square wave

Objective: interface the FPGA to the DAC0800 and generate a square wave.

1. Connect the circuit as shown in Figure 10.1.

2. Connect D0–D7 of the DAC to general-purpose I/O pins on the DE2 board.
3. Write Verilog code to generate a square wave with a frequency of 100 Hz and a 70% duty cycle.
4. You may use a lookup table, following the same approach as the LCD program in Experiment 5.
5. Print and submit the Verilog code and simulation result.

Activity 3 — Generating a sine wave

Objective: interface the FPGA to the DAC0800 and generate a sine wave.

1. Using the equation in the Introduction, complete Table 10.2.
2. Connect the circuit as shown in Figure 10.1.
3. Connect D0–D7 of the DAC to general-purpose I/O pins on the DE2 board.
4. Write Verilog code to generate a sine wave with a frequency of 100 Hz.
5. You may use a lookup table, following the same approach as the LCD program in Experiment 5.
6. Print and submit the Verilog code and simulation result.

Table 10.2: Angle vs. voltage magnitude for the sine wave.

Angle (°)	$\sin\theta$	V_{out} (V)	Value sent to DAC ($V_{out} \times 25.6$)
0			
30			
60			
90			
120			
150			
180			
210			
240			
270			
300			
330			
360			